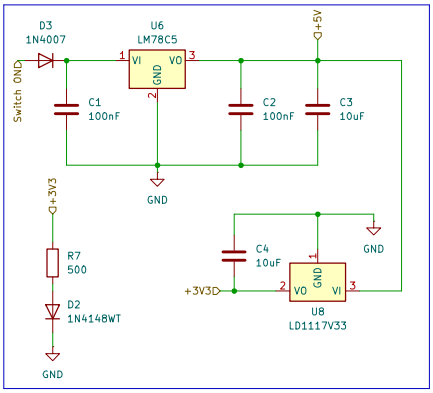
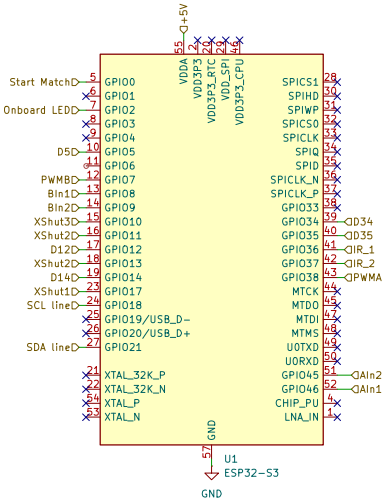
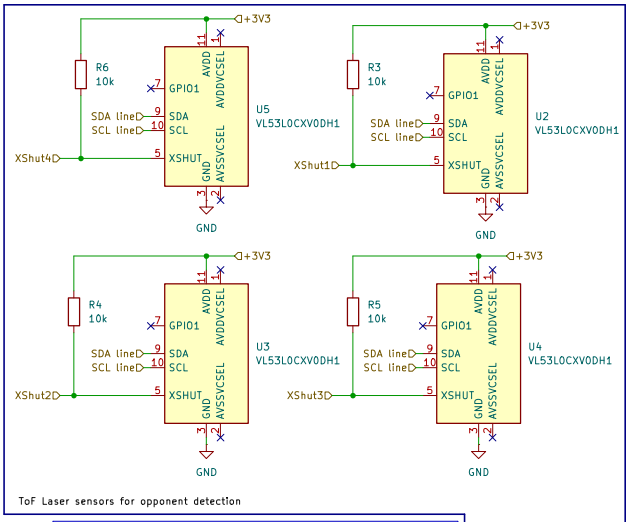
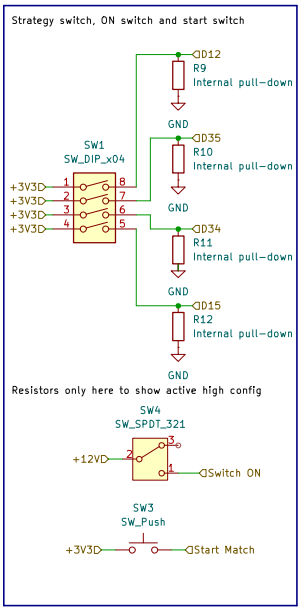
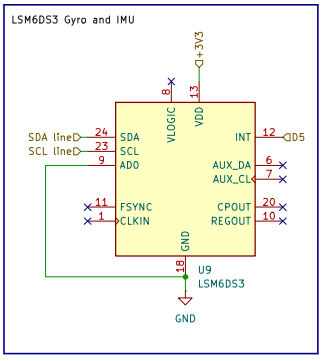
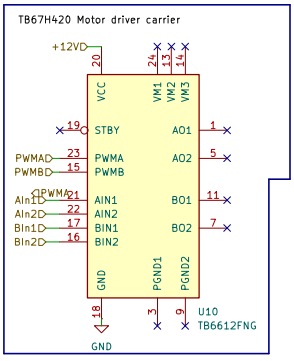

D3
1N4007
U6
LM78C5
Switch ON
C1
100nF
C2
100nF
C3
10uF
+5V
GND
+3V3
R7
500
D2
1N4148WT
C4
10uF
U8
LD1117V33

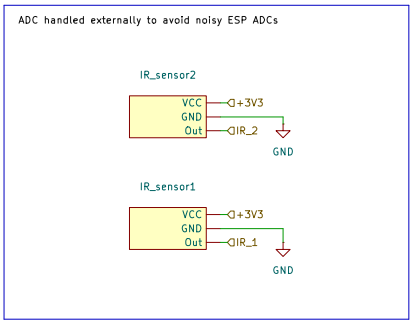

+5V
Start Match
Onboard LED
D5
PWMB
Bin1
Bin2
XShut3
XShut2
D12
XShut2
D14
XShut1
SCL line
SDA line
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO17
GPIO18
GPIO19/USB_D-
GPIO20/USB_D+
GPIO21
XTAL_32K_P
XTAL_32K_N
XTAL_P
XTAL_N
VDDA
VDD3P3
VDD3P3_RTC
VDD_SPI
VDD3P3_CPU
SPICS1
SPIHD
SPIWP
SPICS0
SPICLK
SPIQ
SPID
SPICLK_N
SPICLK_P
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO38
MTCK
MTDO
MTDI
MTMS
U0TXD
U0RXD
GPIO45
GPIO46
CHIP_PU
LNA_IN
D34
D35
IR_1
IR_2
PWMA
AIn2
AIn1
GND
U1
ESP32-S3

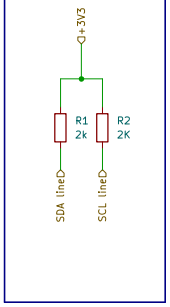

+3V3
R6
10k
R3
10k
R4
10k
R5
10k
GPIO1
SDA line
SCL line
SDA
SCL
XSHUT
AVDD
AVDDVCSEL
AVSSVCSEL
GND
XShut4
XShut1
XShut2
XShut3
U5
VL53L0CXV0DH1
U2
VL53L0CXV0DH1
U3
VL53L0CXV0DH1
U4
VL53L0CXV0DH1
ToF Laser sensors for opponent detection


Strategy switch, ON switch and start switch
SW1
SW_DIP_x04
+3V3
D12
R9
Internal pull-down
D35
R10
D34
R11
D15
R12
GND
Resistors only here to show active high config
SW4
SW_SPDT_321
+12VD
Switch ON
SW3
SW_Push
Start Match


LSM6DS3 Gyro and IMU
+3V3
SDA line
SCL line
SDA
SCL
ADO
VLOGIC
VDD
INT
D5
AUX_DA
AUX_CL
FSYNC
CLKIN
CPOUT
REGOUT
GND
U9
LSM6DS3


TB67H420 Motor driver carrier
+12VD
VCC
STBY
PWMA
PWMB
AIn1
AIn2
BIn1
BIn2
VM1
VM2
VM3
AO1
AO2
BO1
BO2
AIN1
AIN2
BIN1
BIN2
PGND1
PGND2
GND
U10
TB6612FNG


ADC handled externally to avoid noisy ESP ADCs
IR_sensor2
VCC
GND
Out
+3V3
IR_2
GND
IR_sensor1
IR_1


+3V3
R1
2k
R2
2K
SDA line
SCL line